

Turning Churn Data into Protected Revenue

A data-driven retention engine for a US wireless carrier

Proof of Concept • Market analysis • EDA • Machine learning • Business case

Recommendation: a value-based ('Expected Value at Risk') targeted-retention program

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The problem, the idea, and the payoff on one page

- ▶ **PROBLEM** — In a saturated US wireless market (~2% monthly churn, ~22%/yr), the client loses high-value customers faster than it can profitably replace them; winning back a lost customer costs 5–10× more than keeping one.
- ▶ **INSIGHT** — Churn here is not random and not demographic. It is an equipment & engagement problem: aging handsets, no modern (web-capable) devices, low/declining usage, and overage bill-shock are the dominant, fixable drivers.
- ▶ **SOLUTION** — Score every customer monthly, then rank by Expected Value at Risk ($\text{EVaR} = P(\text{churn}) \times \text{revenue}$) and route the top names into four persona-matched offers (device upgrade, right-plan, loyalty, re-engage).

\$1.64M

net benefit / yr
(per 100k customers)

164%

return on the
retention budget

45%

of revenue-at-risk
in just the top 20%

\$32.8M

net / yr scaled to a
2M-subscriber base

A saturated market where retention beats acquisition

- ▶ The US wireless services market is ~\$215–321B with 632M connections, but growing only ~2.4%/yr — a mature, share-of-wallet fight, not a land grab. [1][2]
- ▶ Carriers run ~0.86–0.92% postpaid churn per month (~22%/yr of the base must be re-won every year). [3][4]
- ▶ Acquiring a customer costs 5–10× more than retaining one; lifting retention 5% can raise profit 25–95%. [5][6]
- ▶ Each gross add costs ~\$350 (CPGA); every avoided churn protects 12+ months of revenue and dodges that cost. [7]

\$215–321B

US wireless market size [1][2]

~22% / yr

of customers churn [3][4]

5–10×

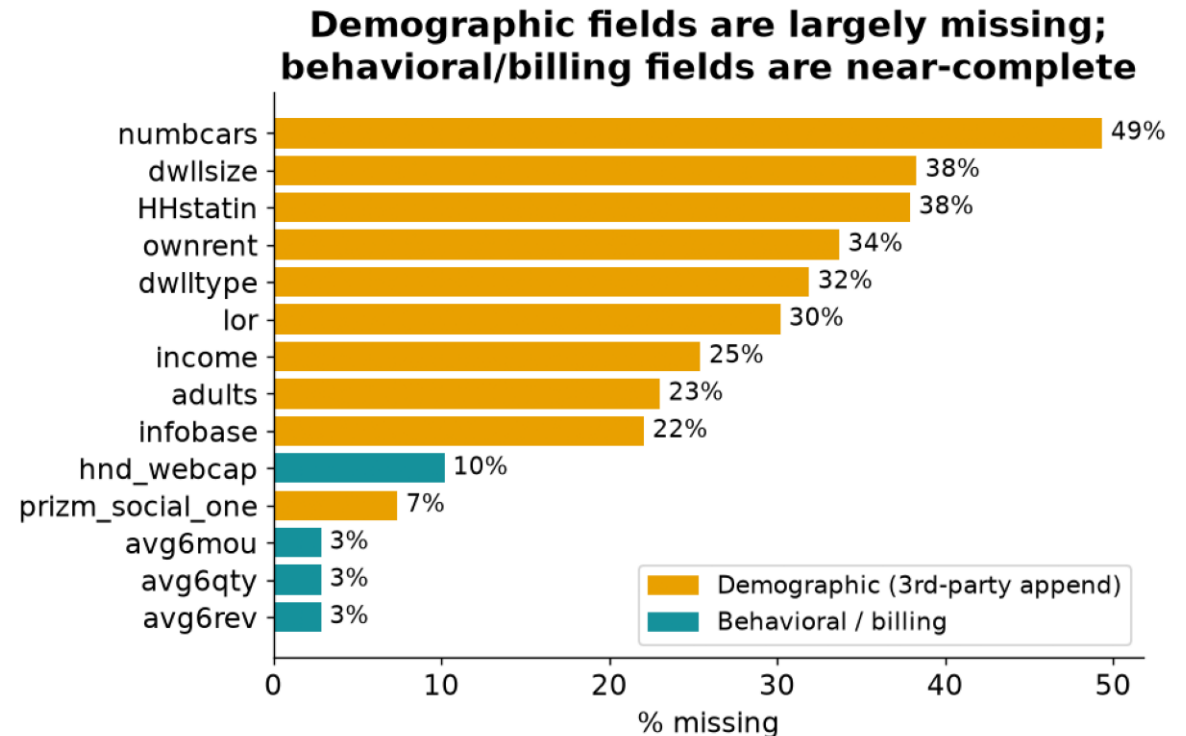
costlier to acquire vs retain [5]

+25–95%

profit from +5% retention [6]

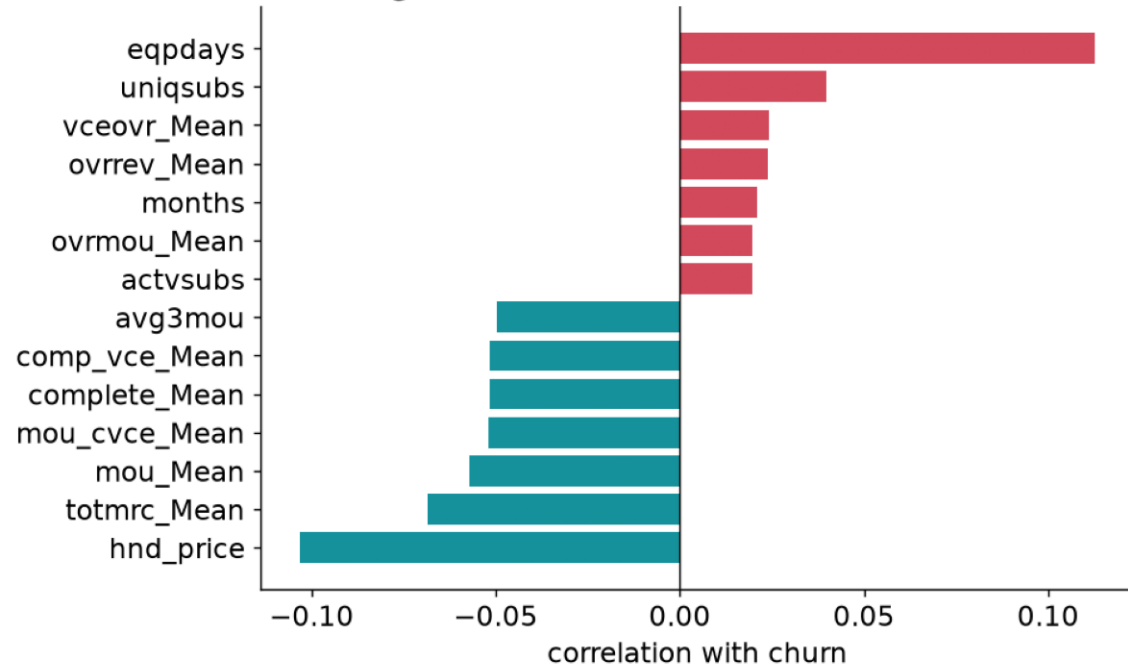
100k customers, 100 fields — but trust behavior, not demographics

- ▶ Merged Client.csv + Record.csv on Customer_ID → 100,000 customers × 100 features (usage, billing, equipment, call-quality, demographics + churn label).
- ▶ Target is balanced (~49.6% churn) — the sample is oversampled for modeling; we re-base to the real ~2%/mo rate in the business case.
- ▶ Third-party demographic fields are 25–49% missing (income, dwelling, #cars). Behavioral/billing fields are ≥99% complete.
- ▶ So the model — and every intervention — is built on reliable behavioral signals, not sparse demographics.

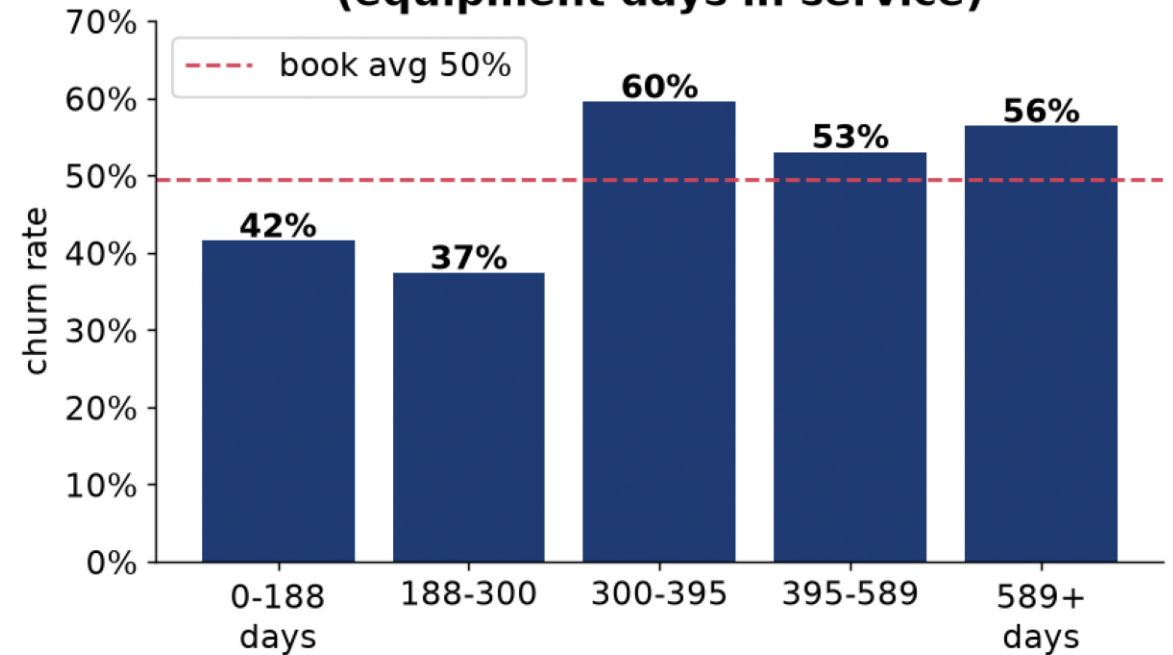


What actually moves churn: equipment age leads

What moves churn: equipment age up, handset price & usage down (Pearson corr. with churn)



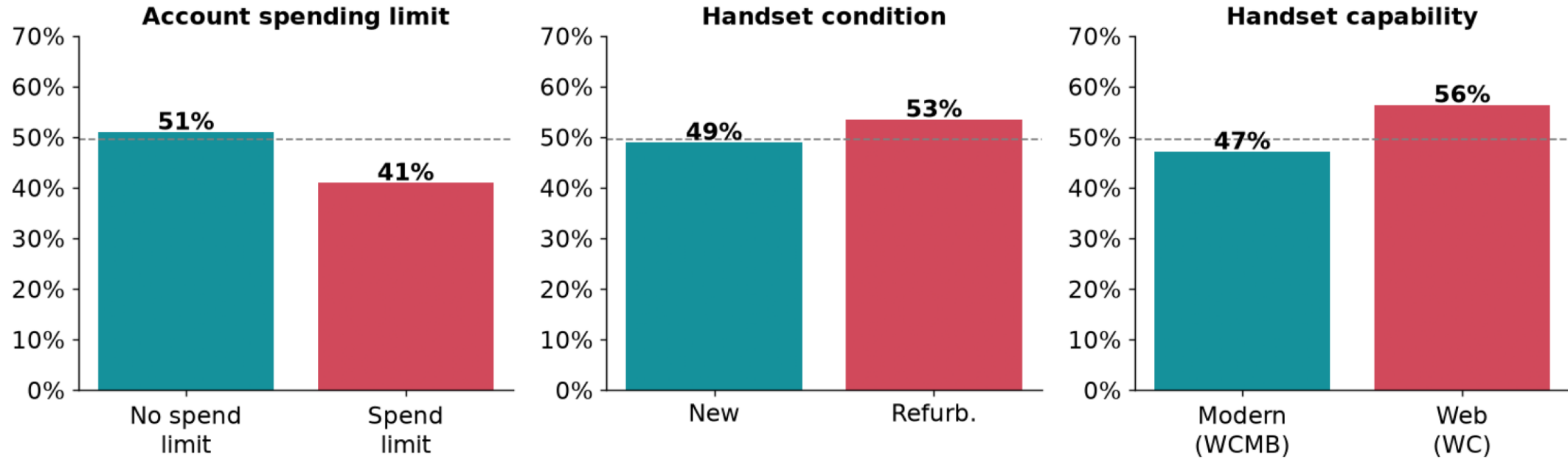
Churn rises with handset age (equipment days in service)



- ▶ Equipment age (eqpdays) is the #1 driver: churn jumps from ~37-42% on newer phones to ~56-60% on older ones.
- ▶ Higher handset price & higher usage (mou) pull churn DOWN — engaged, well-equipped customers stay.

Three interventionable levers stand out

Equipment & account levers separate churners from stayers

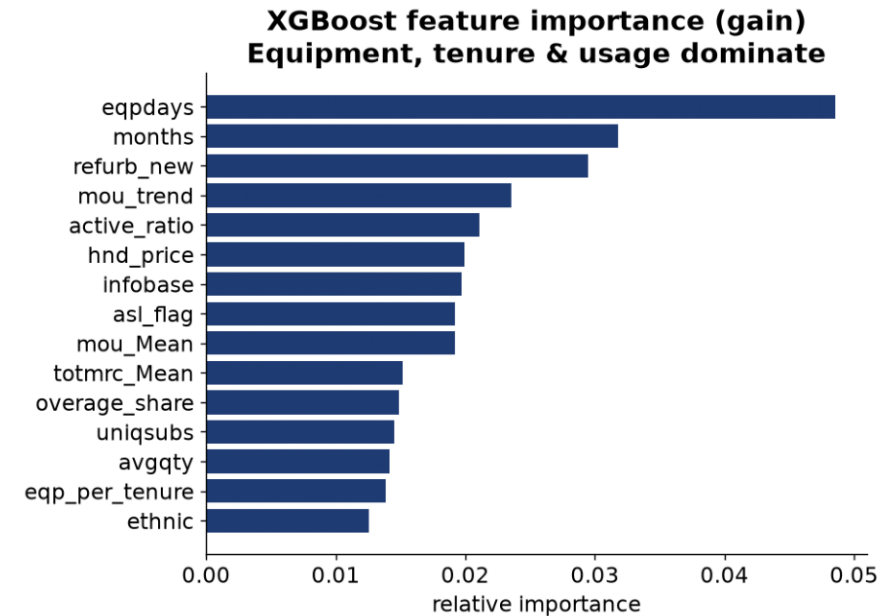
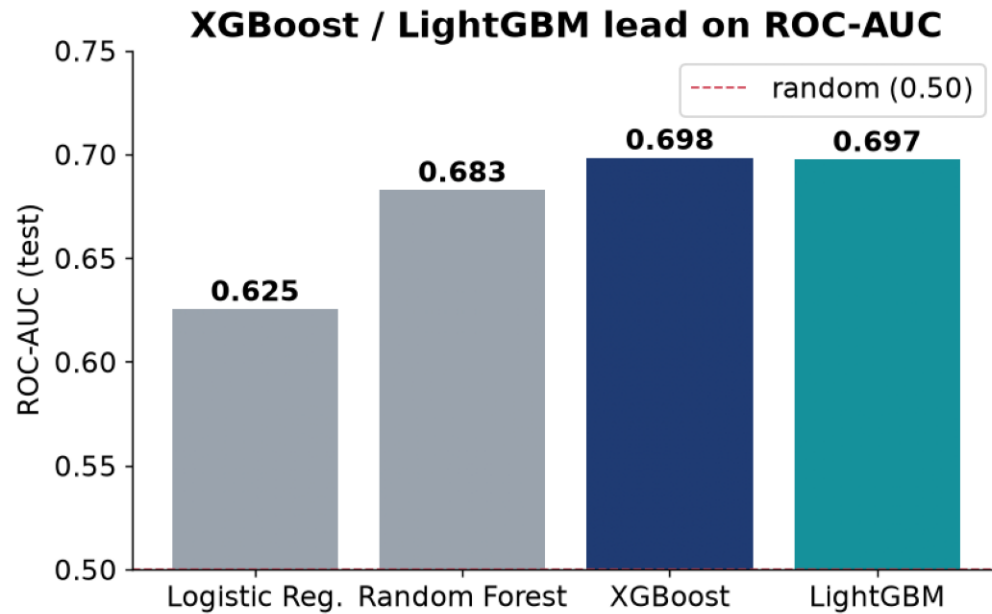


- ▶ Spend-limit accounts (asl_flag=Y) churn ~10 pts LESS — a commitment signal we can nurture.
- ▶ Refurbished handsets churn MORE than new — device quality matters.
- ▶ Phones without modern web capability churn MOST — a direct upgrade trigger.

From a vague 'reduce churn' goal to a precise, profitable decision

- ▶ Business objective — reduce voluntary churn and the revenue it destroys, profitably (not churn for its own sake).
- ▶ ML task — binary classification: predict $P(\text{churn})$ in the 30–60 day window from the labeled churn outcome.
- ▶ Our twist — do NOT stop at probability. Rank by Expected Value at Risk: $\text{EVaR} = P(\text{churn}) \times \text{customer revenue}$, so the budget protects the most REVENUE, not merely the most accounts.
- ▶ Success metrics — ROC-AUC (ranking quality) + lift / capture in the top decile-quintile + revenue-at-risk captured — the numbers that map straight to a campaign.
- ▶ Why this problem — its drivers (equipment, plan-fit, engagement) are observable AND fixable, so a prediction becomes an action.

Gradient boosting wins; the model agrees with the EDA



- ▶ Model & score — we compared 4 models on a held-out 30% test set. Selected model: XGBoost, evaluation metric ROC-AUC = 0.698 (5-fold CV 0.700 ± 0.003); accuracy 0.64, lift@20% 1.49. (Logistic 0.625, Random Forest 0.683, LightGBM 0.697.)
- ▶ Why this model & how built — gradient boosting captures non-linear interactions and handles mixed numeric/categorical fields & missing values without heavy preprocessing; built with median imputation + 5 engineered ratios, 600 trees (depth 5, learning-rate 0.03), validated by stratified 5-fold CV.
- ▶ Insight — top drivers (eqpdays, refurb_new, months, hnd_price, mou) mirror the EDA, so the model is accurate AND explainable; AUC ≈ 0.70 is the realistic ceiling for this classic dataset. [8]

Rank by revenue-at-risk, not just probability

- ▶ A churn-probability list over-weights cheap, low-value accounts.
- ▶ Multiplying probability $\times$ revenue (EVaR) re-prioritises toward customers whose loss actually hurts.

▶ Same campaign size (top 20%), far more value protected.

45%

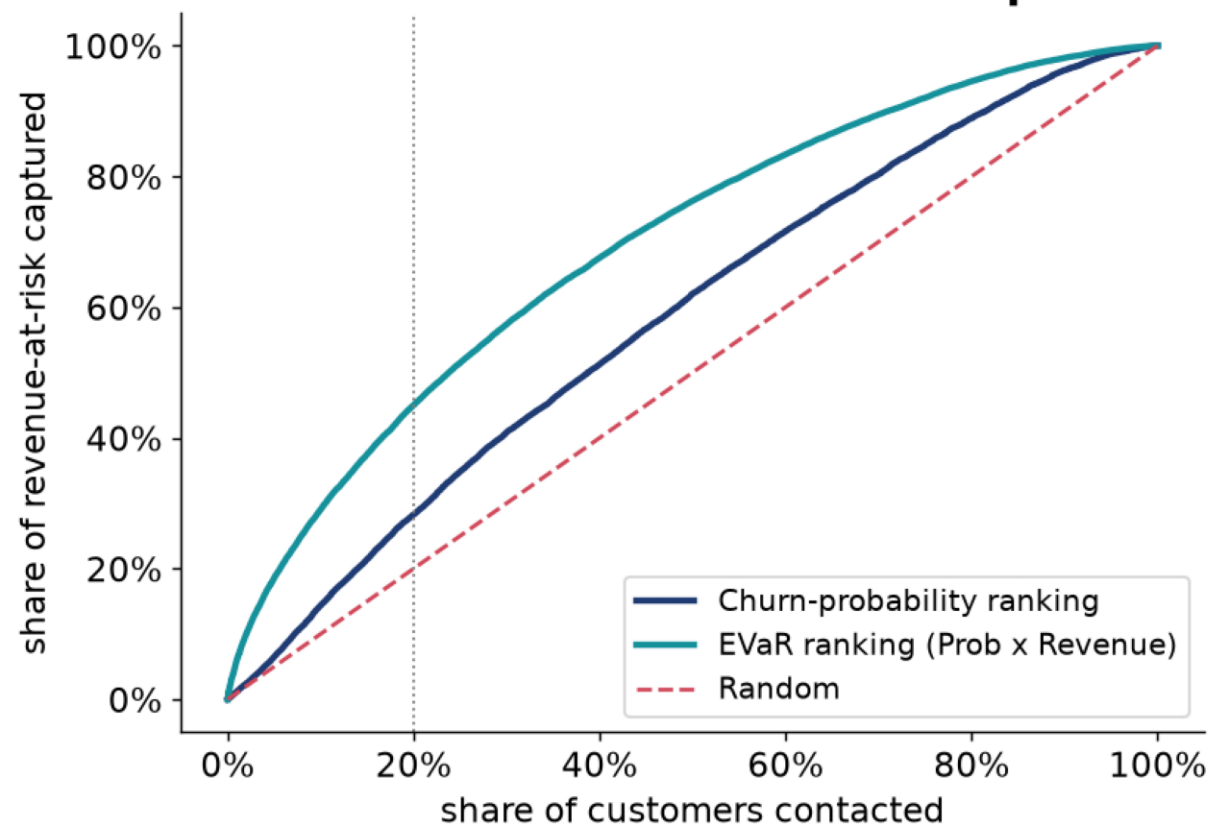
revenue-at-risk
captured (EVaR)

28%

captured by naive
churn ranking

→ **EVaR protects 1.6 $\times$ more revenue than churn-ranking and for the exact same number of customers contacted.**

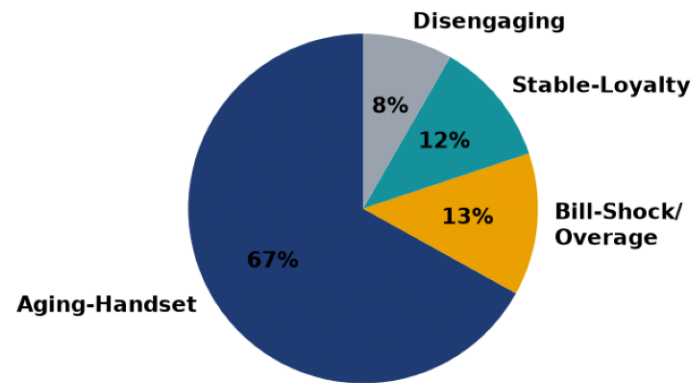
**EVaR targeting protects far more revenue:
45% of revenue-at-risk in the top 20%**



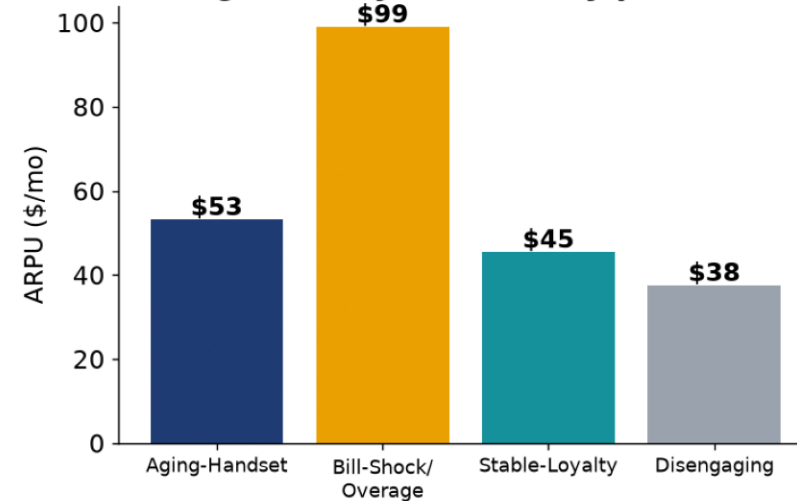
Four at-risk personas, each with a matched offer

Two-thirds of at-risk customers are an aging-handset problem

Composition of the at-risk pool



Avg monthly revenue by persona

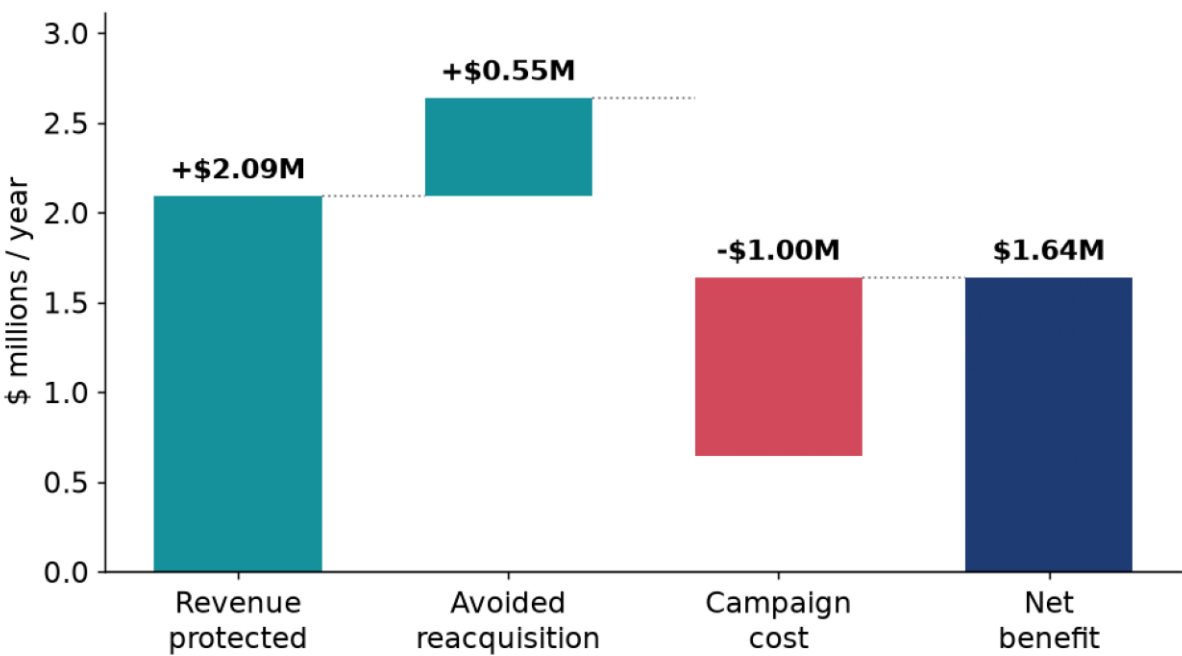


- ▶ Aging-Handset (~67%) → proactive financed DEVICE UPGRADE — the single highest-ROI lever.
- ▶ Bill-Shock / Overage (~13%, ARPU \$99) → RIGHT-PLAN move to unlimited + proactive overage alerts.
- ▶ Stable-Loyalty (~12%) → LOYALTY credit / contract-renewal nudge.
- Disengaging (~8%) → RE-ENGAGE bundle / win-back.

What it earns: ~\$1.64M net per 100k customers / year

- ▶ Per 100k customers, ~\$15.5M of revenue is at risk each year (re-based to ~22%/yr real churn).
- ▶ Target the top 20% by EVaR → reach ~45% of that revenue; a 30% save rate protects ~\$2.09M.
- ▶ Add ~\$0.55M avoided reacquisition cost; subtract ~\$1.0M campaign cost (\$50 × 20k offers).

Retention program economics — per 100k customers / year
Net benefit \$1.64M • ROI 164%



\$1.64M

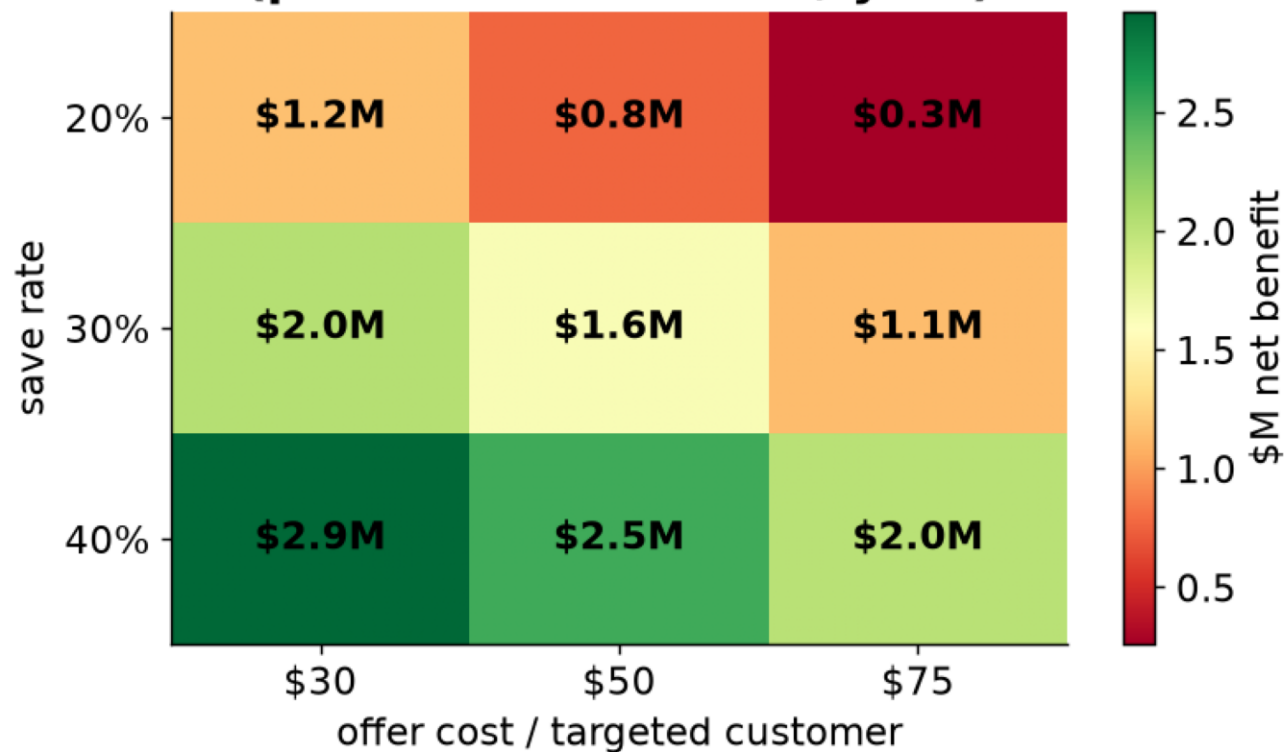
net benefit / yr

164%

ROI on spend

Profitable across every assumption — and it scales

Net benefit stays positive across assumptions (per 100k customers / year)



- ▶ Net benefit stays POSITIVE across every save-rate (20–40%) × offer-cost (\$30–75) scenario tested.
- ▶ Even pessimistic (20% save, \$75 offer) the program nets +\$0.3M per 100k customers.
- ▶ Linear to scale: a 2M-subscriber base implies ~\$32.8M net benefit per year.
- ▶ Only the population churn rate is an external assumption; the model's lift was measured on held-out data.

A staged, measured rollout

90-day pilot

- ▶ Weeks 1-2 — deploy monthly scoring, build EVaR list, assign personas, design offers.
- ▶ Weeks 3-8 — pilot in one region with a randomized CONTROL group to measure true incremental save.
- ▶ Weeks 9-12 — scale winners, tune offer mix per persona, stand up a monthly scoring pipeline.

Risks & mitigations

- ▶ Balanced sample $\neq$ live 2% base $\rightarrow$ recalibrate probabilities before budgeting.
- ▶ Offer cannibalization (paying people who'd stay anyway) $\rightarrow$ control group + uplift modeling (treat the persuadable).
- ▶ Model drift $\rightarrow$ monthly retrain & monitoring; Fairness $\rightarrow$ exclude protected demographics (already weak), audit offers.

Why this proposal wins

- ▶ Churn here is a fixable equipment & engagement problem — the data points straight at the levers.
- ▶ EVaR targeting is the differentiator: it protects 45% of revenue-at-risk from the top 20% — 1.6× a naive churn list.
- ▶ Persona-matched offers turn a score into specific actions; two-thirds of the at-risk pool is one upgrade program.
- ▶ The economics are compelling and robust: ~\$1.64M net / 100k customers (164% ROI), positive in every scenario, ~\$32.8M at 2M scale.
- ▶ Honest & deployable: explainable model, stated assumptions, control-group pilot, fairness & drift safeguards.

Citations

[1] MarketLine (2024). Wireless Telecommunication Services in the United States — Market size & growth.

[2] IBISWorld (2024). Wireless Telecommunications Carriers in the US — Industry report.

[3] T-Mobile US (2025). Q4 2024 results — postpaid phone churn 0.92%.

[4] Verizon Communications (2025). Q4/FY2024 earnings — churn ~0.89%, ARPA \$139.77.

[5] SAS / Harvard Business Review — Cost of acquisition vs retention (5–10×); ~22%/yr attrition.

[6] Bain & Company / Reichheld; Simon-Kucher — +5% retention → +25–95% profit.

[7] Investopedia (2006). Cost Per Gross Add (CPGA) — ~\$350 acquisition cost.

[8] Course tutorial notebook & Cell2Cell telecom churn benchmark (AUC ~0.70 ceiling).

[9] Dataset: Client.csv + Record.csv (provided with assignment), 100,000 customers.

[10] Generative-AI assistance (Devin) — session / chat history:

<https://app.devin.ai/sessions/513c994718b44c95a14cdce101e30bdb>

AI was used for code scaffolding and prose editing; all analysis, metrics and figures are reproduced by the accompanying notebook (Hazem_Douzi.ipynb).